🌞 DayNightSystem - Dalil L-Inspector b Darija

Char7 mufassal l-koul option, variable, w bouton f l-Inspector dial Unity 6 (URP)

MOROCCAN DARIJA (DARIJA B L-7OUROUF LATINIYA) UNITY 6 • ALLSKYFREE 10 SKYBOXES

💡 **Fikra 3amma:** Had l-system msawwb bach idewwer nhar w lil (24 sa3a) b 10 dial les skyboxes AllSkyFree. L-l3ba katbda f **07:00 AM (Sbah)**, w sma katbeddel chwya b chwya (Smooth S-Curve) bla ma t-bloqa w bla ma twlli d-dnya kahla f marra!

🎮 1. Controls l-foqaniyin (Header & Quick Jump Buttons)

Hadou homa l-adawat li kat-chouf f ras l-Inspector bach t-testi l-waqt f Edit Mode bla ma t-l3ab:

`Time (0h - 24h) Slider` — `Slider (0.00 → 24.00)`

Slider dial l-waqt l-7ay f Unity Editor.

Chnou kaydir: Jer had l-curseur b la souris f Edit Mode w ghat-chouf direct f Scene View sma, l-chams, d-dlam, w les couleurs kaytbeddlo f nafs l-waqt. Kaykhellik t-chouf ayy sa3a bghiti bla ma t-dir Play.

`Quick Jump Buttons (Les Boutons d Zerba)` — `Inspector Buttons`

8 dial les boutons li kay-naqzou direct l-awqat mohimma:

- **07:00 (Start):** Kayrd l-l3ba l 7:00 d sbah (l-bidaya dial l-game m3a sma zarqa w chams tal3a).
- **10:00 (Day):** Kayddik l sbah l-mtiyez (Overcast Low daylight).
- **14:30 (Pink):** Golden Afternoon (l-3chiya l-dhbiya m3a sma fiha l-werdi).
- **17:00 (Sunset):** L-maghrib (chams hamra w limouniya tal3a f l-ofsoq).
- **19:30 (Dusk):** Chafaq / Deep Dusk (l-violet w l-indigo).
- **21:30 (Moon):** L-lil m3a l-gmr (Moonlit Night b dow zreq fathi).
- **00:00 (Mid):** Nss l-lil (Cold Midnight starry sky).
- **05:00 (Dawn):** L-fjer l-fada'i (Pre-Dawn Cosmic sky).

⚡ `Auto-Populate 10 AllSkyFree Skyboxes` — `Action Button`

Bouton d l-automatic setup dial ga3 l-10 Skyboxes.

Chnou kaydir: Klik wahed 3lih kaymchi l dossier `Assets/AllSkyFree`, kayjbed ga3 l-10 materials Equirect w les textures Cubemaps, kayrttebhom b l-waqt dialhom (mn 6h d sbah tal l-lil), w kay3tihom les couleurs d chams w l-ambient bla ma t-3mmer chi 7aja b yddik.

⏰ 2. Time Settings (I3dadat L-Waqt)

Hna kat-t7kkem f kifach l-waqt kaydouz w ch-hal kat-stghreq koul 24 sa3a:

`Current Time`

`float (Range 0.0 - 24.0)`

L-waqt l-7ali f l-l3ba b l-a3dad l-3achriya.

> **Chnou kaydir:** Kay-stori sa3a d daba. Matlan `7.0` kat3ni 7:00 AM, `14.5` kat3ni 2:30 PM (14h30), w `21.75` kat3ni 9:45 PM. L-game mprogagrmé kaybda f **7.0** dima.

`Day Duration In Minutes`

`float (Default: 5.0)`

Ch-hal mn dqiqa f l-waqi3 (Real-world minutes) khassha bach idouz nhar kamel f l-l3ba.

> **Chnou kaydir:** Ila derti `5.0`, nhar kamel dial 24 sa3a f l-l3ba ghaydouz f 5 d-dqayeq dial besseh. Ila bghiti nhar itewwel, dir `10.0` (10 dqayeq) wlla `15.0`.

`Time Multiplier`

`float (Default: 1.0)`

Sor3a dial douz l-waqt (Fast-Forward).

> **Chnou kaydir:** `1.0` howa l-3adi. Ila bghiti t-zreb l-waqt f Play Mode bach tchouf l-ghroub w l-lil bzerba, dir `5.0` wlla `20.0`.

`Is Paused`

`bool (Checkbox)`

T-waqqaf l-waqt (Pause).

> **Chnou kaydir:** Ila kochetiha, l-waqt kay-tbloqa f sa3a li raki fiha ma kaye-bqa-ch it7rrek, mziana ila bghiti t-focusi 3la chi mar7ala f l-l3ba.

`Current Day`

`int (Default: 1)`

Nemra dial nhar (Compteur).

> **Chnou kaydir:** Kaybda f Day 1. Koulma katkemmel 24 sa3a w twsel l 00:00, kaytzad nhar jdid automatic (Day 2, Day 3...).

🌞 3. Celestial Lighting (L-Chams w L-Qmer)

Had l-options homa li 7ellou l-mochkil dial "blue then black" bach maykhelliwch l-dnya twlli kahla f marra:

`Sun Light` `Light (Directional Light)`

Dow l-ra'issi dial l-3alam (Directional Light f l-hierarchy).

> **Chnou kaydir:** Kay-attachi Directional Light dial l-scene. Had l-light kayakhoud dawr l-chams b nhar w dawr l-qmer b l-lil.

`Control Sun Rotation` `bool (Checkbox)`

Douran dial l-chams w l-qmer m3a sa3a.

> **Chnou kaydir:** B nhar kayle3 l-chams mn ch-cherq (10°), katwsel l-zenith (75°) f dhohr, w katghreb f l-gherb. B l-lil, kayakhoud dawr l-qmer w **kaybqa dima dow dayer mn l-foq** bach l-ard matwllich kahla!

`Control Sun Light` `bool (Checkbox)`

T-beddal l-couleur w l-intensity dial dow.

> **Chnou kaydir:** Dow kaytbeddel f koul taniya: f sbah kaykoun dhbi byed, f dhohr pure white, f l-ghroub limouni w 7mer, w f l-lil kaywlli zreq fathi hadé (silvery-blue moonlight).

`Control Ambient` `bool (Checkbox)`

Dow l-bî'a l-3amm (Ambient Light).

> **Chnou kaydir:** Kayt7kkem f `RenderSettings.ambientLight`. Muhimm bzaf 7it kay-filli d-dlam dial l-lil b-dow zreq hadé bach l-perso w l-ard ibano w matkounch 3amyak d-dlam.

`Control Fog` `bool (Checkbox)`

D-dbaba l-jawwiya (Atmospheric Fog).

> **Chnou kaydir:** Ila knti m-active Fog f RenderSettings, had l-option katbeddel couleur d-dbaba automatic bach twali f nafs loun l-ghroub wlla loun sma.

`Sun Azimuth Offset` `float (Degrees, e.g. 50°)`

Zawia dial masar l-chams f l-ofsoq.

> **Chnou kaydir:** Kat-wejjah mnin katle3 l-chams w fin katghreb bnisba l-jbel w l-terrain dialek (Compass rotation).

✨ 4. Continuous Skybox Transition (Smooth Blending)

Hna kayn l-mizan l-kbir li kaykhlli sma t-beddel "chwya b chwya":

`Continuous Blending` `bool (Checkbox, Default: True)`

L-crossfade l-mostamirr f ga3 thwani d nhar.

> **Chnou kaydir:** Had l-option katkhlli l-blending shader `Custom/SkyboxBlendedCubemap` idir fondu (dissolve) S-Curve bin les skyboxes. Ma kaynch plateau w ma kaynch freeze: koul taniya katmchi fiha, sma katbeddel chwya b chwya mn zarqa l-werdiya l-limouniya l-zarqa l-layliya!

`GI Update Interval` `float (Seconds, Default: 2.0)`

Waqt d t-te7yid dial Global Illumination w les réflexions.

> **Chnou kaydir:** Koul 2 twani kay-dir update l `DynamicGI.UpdateEnvironment()` bach l-m3adin w l-ma dial Suimono y-3ekso l-loun jdid dial sma bla ma i-tiyye7 les FPS dial l-game.

🌌 5. الجدول dial 10 AllSkyFree Skyboxes

#	Sa3a	Smiya d l-Phase	Skybox Material f AllSkyFree	Mood w L-Couleur
0	06:00 - 09:30	**Morning Sky** (Bidaya @ 07:00)	`Day_BlueSky_Nothing Equirect`	Sbah dhbi, sma zarqa safya, dow nass3 (1.25)
1	09:30 - 13:30	**Daytime Overcast**	`AllSky_Overcast4_Low Equirect`	Dhohr mtiyez, shab baydin, dow 9wi (1.35)
2	13:30 - 16:30	**Golden Afternoon**	`Epic_GloriousPink Equirect`	L-3chiya l-dhbiya w l-werdiya (1.20)
3	16:30 - 18:00	**Warm Sunset**	`Cold Sunset Equirect`	L-ghroub l-limouni w l-dhbi (0.95)
4	18:00 - 19:15	**Late Sunset**	`Epic_BlueSunset Equirect`	Chafaq 7mer w limouni m3a zreq ghameq (0.70)
5	19:15 - 20:45	**Deep Dusk**	`Deep Dusk Equirect`	Dusk l-violet w l-indigo (0.45)
6	20:45 - 23:30	**Moonlit Night**	`AllSky_Night_MoonBurst Equirect`	L-gmr medwweya w shab feddiyin (0.32)
7	23:30 - 02:00	**Deep Midnight**	`Cold Night Equirect`	Nss l-lil njoum w broud (0.26)
8	02:00 - 04:30	**Late Night**	`Cartoon Base NightSky Equirect`	L-lil l-hadé k7el w zreq ghameq (0.22)
9	04:30 - 06:00	**Pre-Dawn Cosmic**	`AllSky_Space_AnotherPlanet Equirect`	L-fjer l-fada'i violet w rose qbel ch-chrouq (0.42)

6. CaveAtmosphereSystem (Ida'at L-Ghar L-Dakhili)

Hada howa l-system li m-attachi f `CaveAtmosphereSystem` bach dakhil l-ghar maybqa-ch k7el:

`Cave Fill Light` — `Light (Point Light)`

Dow dakhili li kaydwi l-ghorfa l-kbira dial l-ghar.

Chnou kaydir: Kaydwi l-7yout dial l-7ejra w les outils bla ma i-dir ombres tqal. Kayakhoud l-couleur dial l-chams outdoor w kay-synkroh m3a dakhel.

`Day Intensity vs Night Intensity` — `float (Day: 0.65 | Night: 0.08)`

Farq dial dow bin n-nhar w l-lil dakhil l-ghar.

Chnou kaydir: B nhar (07:00 - 18:30) l-ghar kaybqa medwi b `0.65` bach tchouf l-ahjar w l-m3adin mzian. B l-lil kayhbet automatic l `0.08` w kaywlli b loun l-slate/indigo bach i-7ess l-la3eb b-dlam w l-mood l-waqi3i.

`Cave Color Multiplier` — `Color (Warm Subterranean Tint)`

Tezian l-loun dakhil l-ghar.

Chnou kaydir: Kayched l-loun d chams outdoor w kayzid fih chwya d def'e. F l-ghroub, l-7yout d l-ghar kayban fihom le3kas hamr w limouni zwin!

`Cave Entrance Trigger & PostProcess Volume` — `Trigger Zone & Volume Profile`

Dkhla d l-ghar w t-keiyof dial l-3ayn (Eye Adaptation).

Chnou kaydir: Mnin l-perso kaydkhoul mn l-bab d l-ghar, l-trigger kayhsseb l-3umq b-dabd. L-Volume `CaveVolumeProfile` kaytl3 l-exposition (+0.45 EV) bhal l-3ayn dial bnadm mnin katwesse3 f d-dlam, w kay-raf3 l-crevices bach d-dlam maykounch 3amyak.

7. Nassa'i7 d Test w L-Isti3mal

- **Bach t-testi l-waqt:** Mchi l `DayNightSystem` f l-Hierarchy, w 7errek l-slider `Time (0h - 24h)` b chwya b chwya. Ghat-chouf l-transition smooth 100%.
- **Bach t-zreb l-waqt f Play:** Berek 3la l-widget dial sa3a f l-foq 3la limn (wlla wrrek 3la touche `T` f l-clavier), w khtar `5x` wlla `20x`.
- **Ila bghiti nhar itewwel:** Beddel `Day Duration In Minutes` mn 5.0 l 10.0 wlla 15.0 dqiqa.